Lecture 17: Minimum Spanning Trees

Michael Dinitz

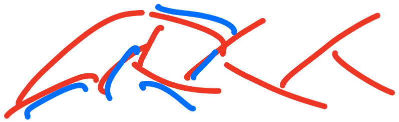
November 3, 2026
601.433/633 Introduction to Algorithms

Announcements

Oral presentations for HW7!

- Must be groups of 3 unless you talked to us about an exception (rarely granted).
- Sign up by Wed 5pm.
- Same system as for HW2 (Canvas). See Courselore.

Introduction



Definition

A *spanning tree* of an undirected graph $G = (V, E)$ is a set of edges $T \subseteq E$ such that (V, T) is connected and acyclic.

Definition

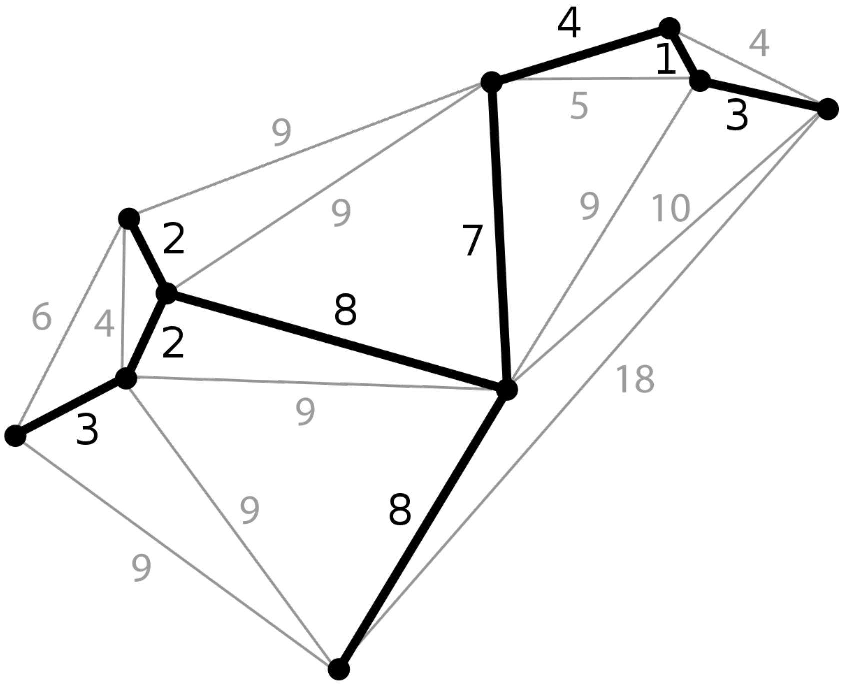
Minimum Spanning Tree problem (MST):

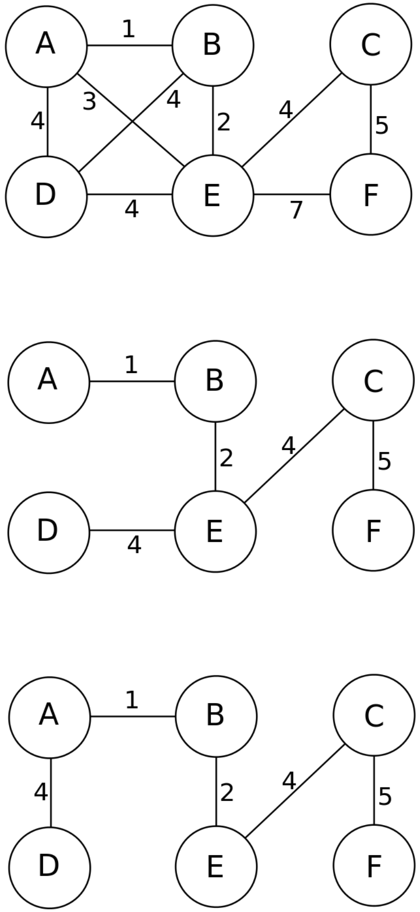
- Input:
  - Undirected graph $G = (V, E)$
  - Edge weights $w : E \to \mathbb{R}_{\geq 0}$
- Output: Spanning tree minimizing $w(T) = \sum_{e \in T} w(e)$.

Foundational problem in *network design*. Tons of applications.

Today: one “recipe”, two different algorithms from recipe. Main idea: greedy.

Examples





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Generic Algorithm

Generic Greedy

Definition

Suppose that A is subset of *some* MST. If $A \cup \{e\}$ is also a subset of some MST, then e is *safe* for A.

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Generic-MST {
    $A = \varnothing$
    while($A$ not a spanning tree) {
        find an edge $e$ safe for $A$
        $A = A \cup \{e\}$
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}
```

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Theorem

Generic-MST is correct: it always returns an MST.

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Generic-MST {
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    while($\boldsymbol{A}$ not a spanning tree) {
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Induction.

Claim: $\boldsymbol{A}$ always a subset of some MST.
Base case: ✓
Inductive step: ✓ □

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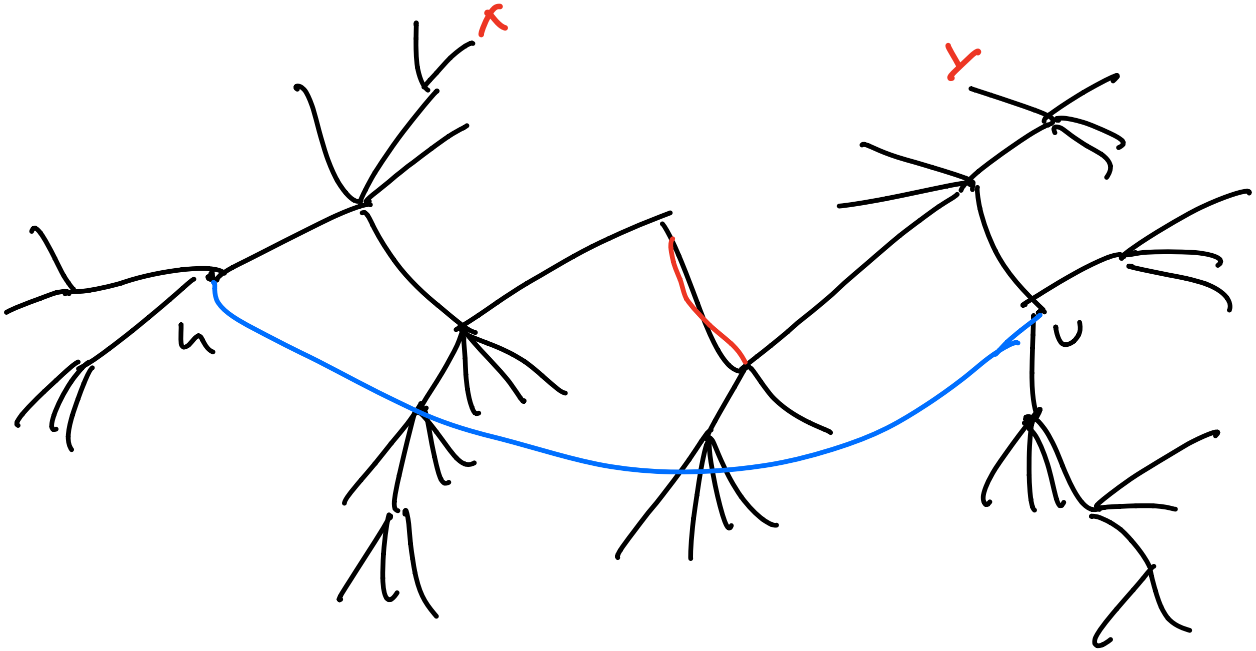
But how to find a safe edge? And which one to add?

Structural Properties

Lemma

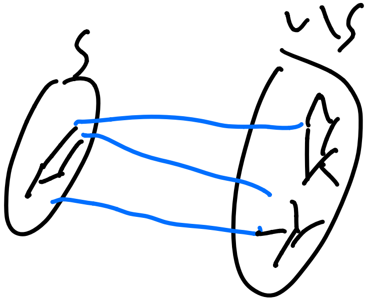
Let $\boldsymbol{T}$ *be a spanning tree, let* $\boldsymbol{u}, \boldsymbol{v} \in \boldsymbol{V}$*, and let* $\boldsymbol{P}$ *be the* $\boldsymbol{u} - \boldsymbol{v}$ *path in* $\boldsymbol{T}$*. If* $\{\boldsymbol{u}, \boldsymbol{v}\} \notin \boldsymbol{T}$*, then* $\boldsymbol{T}' = (\boldsymbol{T} \cup \{\{\boldsymbol{u}, \boldsymbol{v}\}\}) \setminus \{\boldsymbol{e}\}$ *is a spanning tree for all* $\boldsymbol{e} \in \boldsymbol{P}$*.*

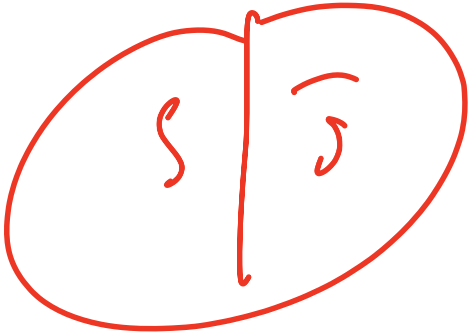


Structural Properties

Definition

A *cut* $(S, V \setminus S)$ (or $(S, \bar{S})$ or just S) is a partition of V into two parts. Edge e *crosses* cut $(S, \bar{S})$ if e has one endpoint in S and one endpoint in $\bar{S}$.





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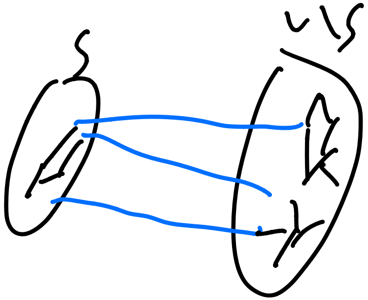
Cut $(S, \bar{S})$ *respects* $A \subseteq E$ if no edge in A crosses $(S, \bar{S})$



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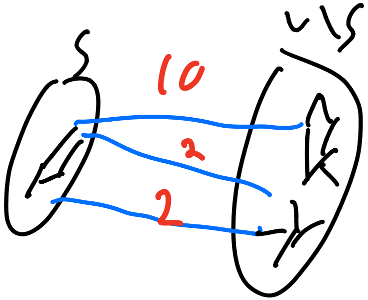

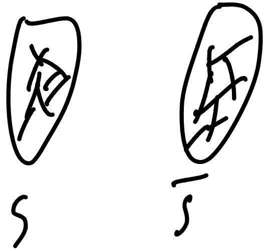

Definition

e is a *light edge* for $(S, \bar{S})$ if e crosses $(S, \bar{S})$ and $w(e) = \min_{e' \text{ crossing } (S,\bar{S})} w(e')$

Main Structural Theorem

Theorem

Let $\boldsymbol{A} \subseteq \boldsymbol{E}$ *be a subset of some MST* $\boldsymbol{T}$, *let* $(\boldsymbol{S}, \bar{\boldsymbol{S}})$ *be a cut respecting* $\boldsymbol{A}$, *and let* $\boldsymbol{e} = \{\boldsymbol{u}, \boldsymbol{v}\}$ *be a light edge for* $(\boldsymbol{S}, \bar{\boldsymbol{S}})$. *Then* $\boldsymbol{e}$ *is safe for* $\boldsymbol{A}$.

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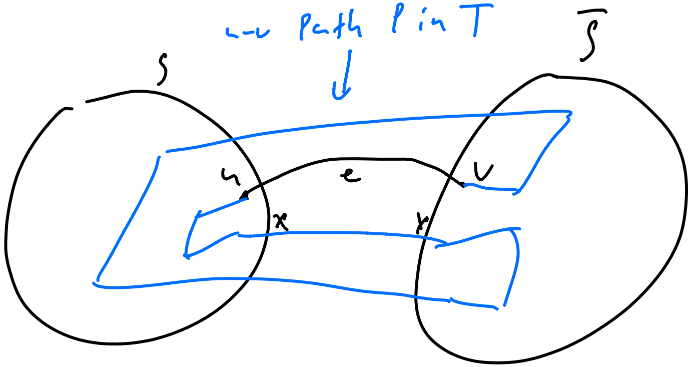
P = u-v path in T

Let $\boldsymbol{T}' = (\boldsymbol{T} \cup \{\boldsymbol{e}\}) \setminus \{\{\boldsymbol{x}, \boldsymbol{y}\}\}$
$\Longrightarrow$ $\boldsymbol{T}'$ a spanning tree by first lemma



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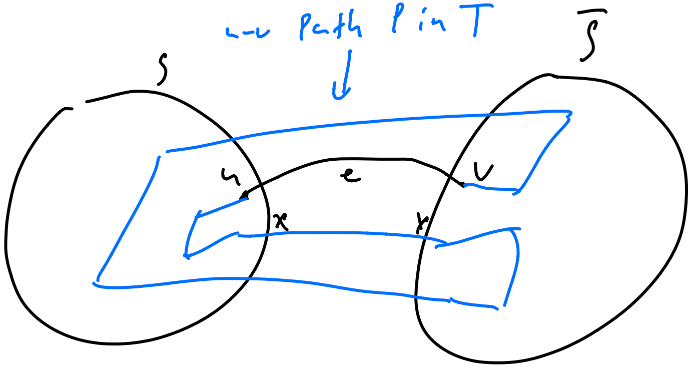

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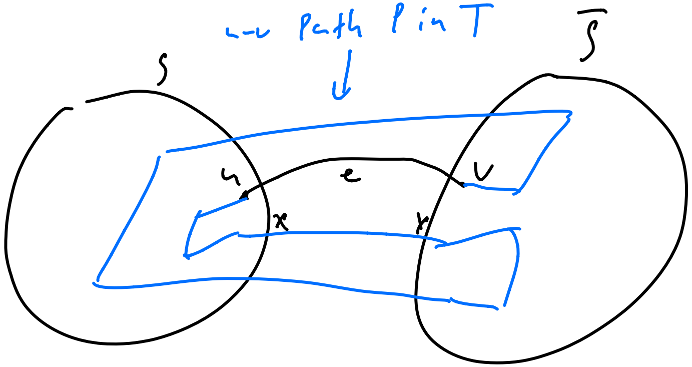
$$\boldsymbol{w}(\boldsymbol{T}') = \boldsymbol{w}(\boldsymbol{T}) + \boldsymbol{w}(\boldsymbol{e}) - \boldsymbol{w}(\{\boldsymbol{x}, \boldsymbol{y}\}) \leq \boldsymbol{w}(\boldsymbol{T})$$



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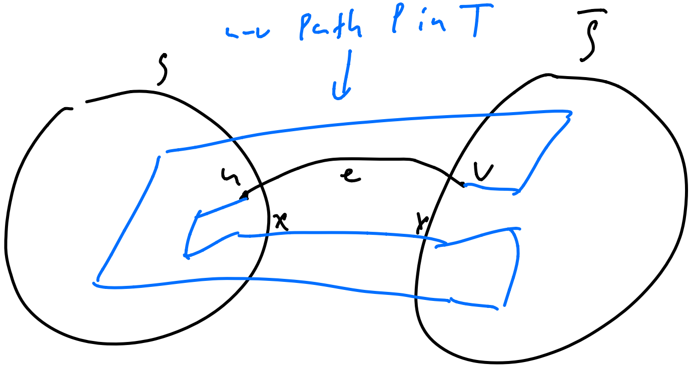

Prim's Algorithm

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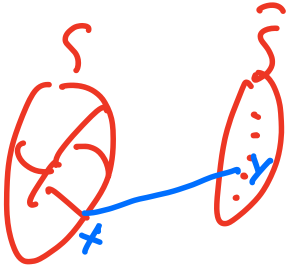

Idea: start at arbitrary node $\boldsymbol{u}$. Greedily grow MST out of $\boldsymbol{u}$.

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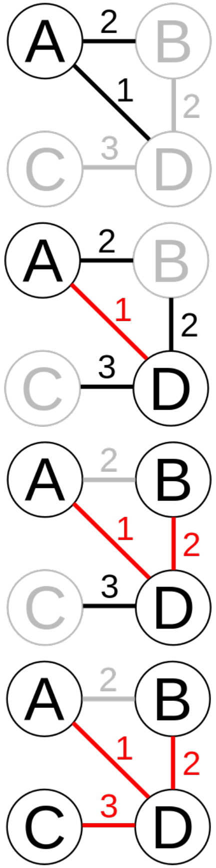
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Proof.

Just Generic-MST!

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Just Generic-MST!

- $(S, \bar{S})$ always respects A (induction).
- If edge e added then light for $(S, \bar{S})$
- Hence e safe for A by main structural theorem.

□

Running Time

Trivial analysis:

- Every spanning tree has $n-1$ edges $\implies$ $n-1$ iterations
- In each iteration, look through all edges to find min-weight edge crossing $(S, \bar{S})$ $\implies$ $O(m)$ time
- Total $O(mn)$

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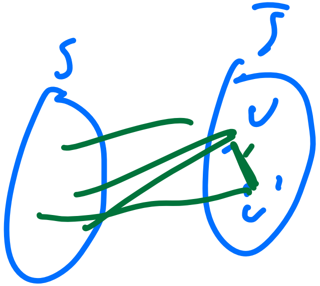

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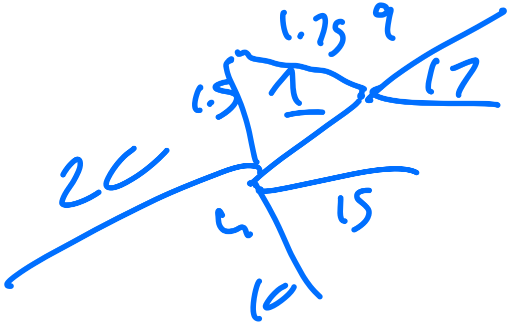
- n Inserts, n Extract-Mins, m Decrease-Keys
- Like Dijkstra, $O(m \log n)$ using binary heap. $O(m + n \log n)$ with Fibonacci heap (only Extract-Min is logarithmic)

Kruskal's Algorithm

Algorithm

Intuition: can we be *even greedier* than Prim's Algorithm?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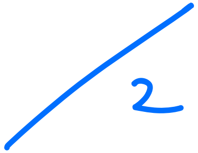

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```
$A = \varnothing$
Sort edges by weight (small to large)
For each edge $e$ in this order {
    if $A \cup \{e\}$ has no cycles, $A = A \cup \{e\}$
}
return $A$
```

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Theorem

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Want to show just Generic-MST: when $\{u, v\}$ added, it was safe for A.

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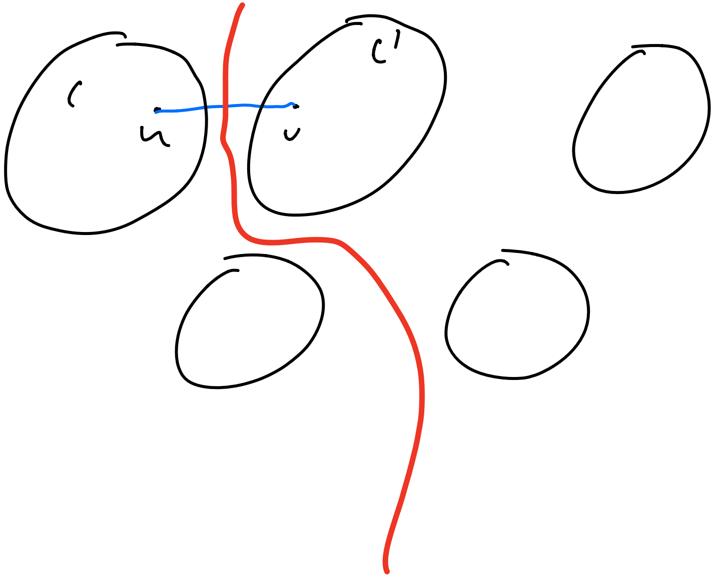

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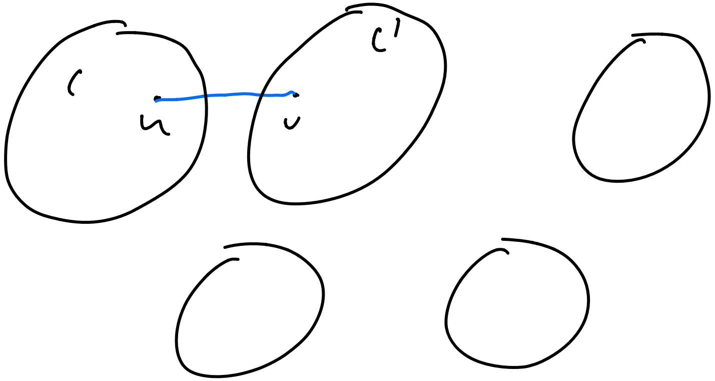

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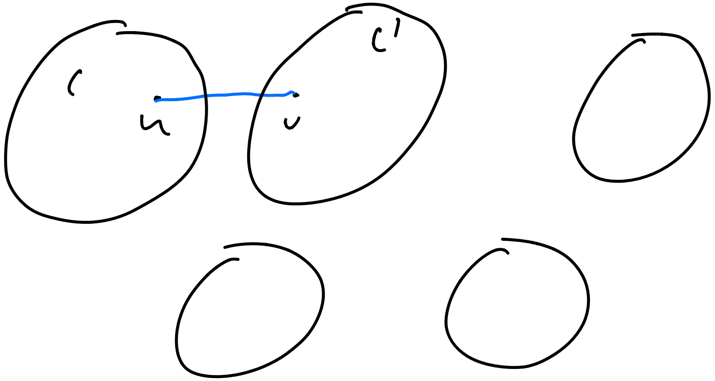

Consider cut $(\boldsymbol{C}, \bar{\boldsymbol{C}})$. Respects $\boldsymbol{A}$, and $\{\boldsymbol{u}, \boldsymbol{v}\}$ light for it.
Main structural theorem $\implies$ $\{\boldsymbol{u}, \boldsymbol{v}\}$ safe for $\boldsymbol{A}$

Running Time

Sorting edges: $\boldsymbol{O(m \log m) = O(m \log n)}$

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Can we speed this up with data structures?

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Can we speed this up with data structures?

Union-Find! Connected components of A are disjoint sets.

Running Time

Sorting edges: $O(m \log m) = O(m \log n)$

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$O(m \log^* n)$ using union-by-rank + path compression

$O(m + n \log n)$ using list data structure

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Sorting dominates! $O(m \log n)$ total.